

■ Plants Need Sunlight

Light Experiment — Detailed Lesson Plan

Grade 2 · Science · Duration: 45–60 min

■ Subject	Science (EVS)	■ Grade	Grade 2
■ Duration	45 – 60 minutes	■ Type	Hands-on Experiment
■ Topic	Plants & Sunlight	■ Setting	Classroom / Home

This lesson plan supports the Grade 2 Science curriculum on living things and their needs. Students explore how plants depend on sunlight through a structured inquiry experiment, vocabulary building, and guided observation activities.

■ Learning Objectives

By the end of this lesson, students will be able to:

- Explain why plants need sunlight to grow.
- Describe what happens to a plant that does not receive sunlight.
- Design a simple controlled experiment comparing light vs. dark conditions.
- Observe and record changes in plants over one week.
- Use science vocabulary: sunlight, experiment, observe, healthy, weak.
- Compare results and draw a conclusion from the experiment.

■ Prior Knowledge Required

- Students should know that plants are living things.
- Basic understanding that living things need certain things to survive (water, food, air).
- Familiarity with simple classroom observation activities.

■ Materials Needed

Item	Quantity	Notes
Moong seeds (mung beans)	10–12 seeds	Available at any grocery store
Small pots or cups	2 per group	Clay or plastic pots work well
Potting soil	1 bag per class	Garden soil also works
Water & watering can	1 per group	Small cup is fine
Cardboard box (dark)	1 per group	Shoebox or large box with lid
Sticky labels / marker	1 set	To label Pot A and Pot B
Observation worksheet	1 per student	See page 5 of this lesson
Coloured pencils / crayons	1 set per student	For drawing observations

■ Lesson Plan & Timing

Phase 1 Warm-Up / Hook

■ 5 min

Show the two potted plant illustrations (one green, one yellow). Ask: "Which plant looks healthy? Why do you think so?" Let students share ideas freely — do not give answers yet. Write key words (sunlight, dark, healthy, weak) on the board.

Phase 2 Concept Introduction

■ 10 min

Introduce the idea that plants are living things that need sunlight to make energy. Draw a simple analogy: "Sunlight is like food for plants, just as rice and lentils are food for us!" Ask students to think of plants they see at home — tulsi, money plant — and where they are kept. Discuss why tulsi is placed near sunny windows.

Phase 3 Experiment Setup

■ 15 min

Step 1 – Plant the Seeds: Students plant moong seeds in two pots with equal amounts of soil. Water both pots with the same amount of water. Label them Pot A (Sunlight) and Pot B (Dark). Step 2 – Place the Pots: Pot A goes near the window. Pot B goes inside the cardboard dark box. Explain: "We are keeping everything the same EXCEPT sunlight — that is our variable."

Phase 4 Observation & Recording

■ 10 min

Distribute observation worksheets. Students predict what will happen to each pot after 7 days and draw their prediction. Teacher introduces vocabulary flashcards: sunlight, experiment, observe, healthy, weak. Students repeat each word and use it in a sentence.

Phase 5 Wrap-Up & Quiz

■ 10 min

Discuss quiz questions as a class: (1) What do plants need to grow? (2) What happens to plants without sunlight? (3) Which pot will grow better? Summarise key learning: Plants need sunlight to grow healthy and strong. Without sunlight, plants become weak and droopy. Plants grow towards light to get energy. Assign home experiment follow-up.

■ Experiment: Step-by-Step Guide

Scientific Question: Does a plant grow better with sunlight or without sunlight?

Hypothesis (Prediction): The plant kept in sunlight will grow taller, greener, and stronger than the plant kept in the dark.

Variable tested: Amount of sunlight received.

Controlled variables: Type of seed, amount of water, type of soil, pot size, room temperature.

■ Plant the Seeds

- Fill both pots with the same amount of soil.
- Push 2–3 moong seeds about 1 cm deep into each pot.
- Gently cover the seeds with soil.
- Water both pots with the same amount of water (e.g. half a cup each).
- Label Pot A with a sticky tag: "Sunlight ■" and Pot B: "Dark ■".

■ Place the Pots

- Put Pot A on a windowsill or near a window that gets at least 4–6 hours of sunlight daily.
- Put Pot B inside a closed cardboard box (or dark cupboard) — no light should enter.
- Both pots should be kept at room temperature.
- Water both pots equally every day at the same time.

■ Observe and Record

- Check both plants every day for 7 days.
- Record observations on your worksheet: leaf colour, height, how upright or droopy the stem is.
- Draw pictures of both plants on Day 1, Day 4, and Day 7.
- Note: Pot A (sunlight) should show green leaves and upright growth by Day 5–7.
- Note: Pot B (dark) will show pale/yellow leaves and a droopy, weak stem.

■ Results & Conclusion

- After one week, compare the two plants side by side.
- Fill in the results table on your observation worksheet.
- Write your conclusion: "Plants need sunlight to grow healthy and strong."
- Discuss: Which plant would you rather be? Why do plants grow toward windows?

■ Student Observation Worksheet

Name: _____ Date: _____ Class: _____

My Prediction (Before the Experiment)

I _____ think _____ Pot _____ A _____ (Sunlight) _____ will: _____

I _____ think _____ Pot _____ B _____ (Dark) _____ will: _____

Daily Observation Record

Day	Pot A – Sunlight (Leaf colour, height, look)	Pot B – Dark (Leaf colour, height, look)
Day 1		
Day 2		
Day 3		
Day 4		
Day 5		
Day 6		
Day 7		

My Conclusion (After the Experiment)

After 7 days, I noticed that Pot A (Sunlight) looked: _____

After 7 days, I noticed that Pot B (Dark) looked: _____

This _____ experiment _____ shows _____ that _____ plants _____ need _____ to grow.

■ Vocabulary Flashcards

Sunlight

■ The light that comes from the sun.
Plants use sunlight to make energy and food.

Experiment

■ A test we do to find out if something is true. We change one thing and watch what happens.

Observe

■ To look carefully at something and notice details. Scientists observe to learn.

Healthy

■ Strong and growing well. A healthy plant has green leaves, a straight stem, and grows tall.

Weak

■ Not strong. A weak plant has yellow leaves, a droopy stem, and does not grow well.

■■ Indian Context — Real-Life Examples

■ Tulsi (Holy Basil)

The tulsi plant is found in almost every Indian home. It is traditionally kept in the courtyard or near a window that gets morning sunlight. Ask students: "Have you seen a tulsi plant at home? Where is it kept? Why do you think grandparents place it in the sun?"

■ Money Plant (Pothos)

Money plants are commonly seen leaning or growing toward the nearest window. This is called phototropism — plants grow toward light. Students can observe this at home and bring their observations to class.

■■ Sunflower Fields

Sunflowers famously turn to face the sun throughout the day (heliotropism). They are a vivid example of how important sunlight is to plant growth and health.

■ Quiz — Let's Check!

Answer the following questions in complete sentences:

Q1. What do plants need to grow?

Answer: Plants need sunlight, water, soil, and air to grow.

Q2. What happens to a plant that does not get sunlight?

Answer: Without sunlight, a plant becomes weak, its leaves turn yellow, and its stem droops.

Q3. Which pot grew better in our experiment — Pot A or Pot B? Why?

Answer: Pot A (kept in sunlight) grew better because it received sunlight, which plants need to make energy.

Q4. What is the word for looking carefully at something?

Answer: The word is "observe."

Q5. Name two plants from your home that need sunlight.

Answer: Examples: tulsi plant, money plant, aloe vera, coriander (answers may vary).

■ Lesson Summary

- ✓ Plants are living things and need sunlight, water, soil, and air to grow.
 - ✓ Sunlight helps plants make energy — it acts like food for plants.
 - ✓ A plant kept in sunlight will be tall, green, and strong.
 - ✓ A plant kept in the dark will be short, yellow, and weak.
 - ✓ Plants grow toward light — this is called phototropism.
 - ✓ We can test this with a simple experiment using two pots (Pot A and Pot B).
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■■■ Teacher Notes & Differentiation Tips

For early finishers:

Ask them to write a short story from the perspective of Pot B — "I am Pot B and I am not feeling well..."

For struggling learners:

Pair with a buddy. Use picture cards for vocabulary. Let them draw rather than write their observations.

Extension activity:

Introduce the word "photosynthesis" as a bonus science word. Older students can research what chlorophyll does.

Safety note:

Ensure seeds are not put in mouths. Wash hands after handling soil.

Assessment:

Collect observation worksheets on Day 7. Use the conclusion box to assess understanding of cause and effect.

■ Home Assignment

Try this experiment at home! Plant a moong seed in a small cup near a window and another in a cupboard. Water them every day and observe for 5–7 days. Draw what you see and bring your drawings to class. Ask a family member: where do the plants in your home get sunlight?